30/11/23

Probability

Experimental Probability

$P(E) = \frac{\text{No. of trials in event happend}}{\text{Total no. of Trials}}$

Theoratical Probability

$P(E) = \frac{\text{no. of favourable outcomes}}{\text{Total no. of outcomes}}$

For eg → $\frac{\text{No. of heads}}{\text{Total no. of heads}} \rightarrow \frac{117}{200}$

Definition: Probability means possibility with the likelihood or chance of an event occurring. It is expressed as 0 and 1.

Vocabulary:

1. Random Experiment: Procedure or Observation that procedures a definite outcome that ~~out~~ cannot be pedicted with certainity before the experiment is condicted.
2. Event: Specific outcome or its a collection from an experiment to which a probability is assigned.
3. Equally likely events: Events that have the same chance of occurring.
4. Sample Space: The set of all possible outcomes of a random experiment.

Content in the chapter 14

1) Rolling ~~a~~ Die → 1 Die, 2 Die
2) Playing Cards

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3) Tossing Coin → 1 coin, 2 coin, 3 coin

4) Numbers → ODD/Even, Composite/Prime, natural no./Integers

5 Mixed bag / Miscellaneous → Real life examples

* Probability of an event is denoted by P(E)
* Sure Event → chances = 100%
  For eg: P(Sun rises in East) = 100% = 100/100 = 1
* Impossible Event: chances = 0.7%
  For eg: P(No. 7 on a dice) = 0% = 0
* Range of Probability =
  $0 \leq P(E) \leq 1$

Case I Rolling Dice:

1 Die: Sample space = {1, 2, 3, 4, 5, 6}
Total no. of outcomes = 6

$P(E) = \frac{\text{No. of favourable outcomes}}{\text{Total outcomes}}$

P(Number less than 2 on die) = 1/6

P(E) = 1/6

2 Die: Sample space:

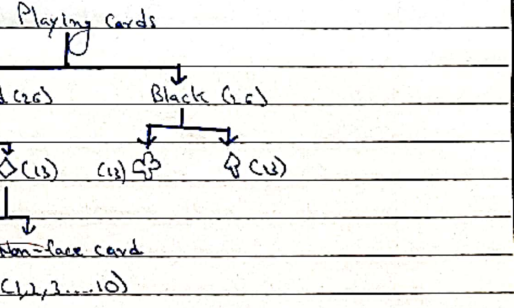
	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Total no. of outcomes = 6 × 6 = 36
Doublets: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6)

Case II Tossing Coin:

1 coin: Sample space: (H, T)
2 coin: Sample space: (HT, TH, TT, HH)
3 coin: Sample space: (HTT, HTH, HHT, HHH, TTT, TTH, THT, THH)

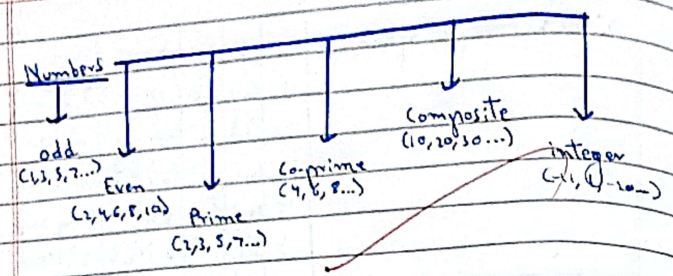
Case III Playing cards:



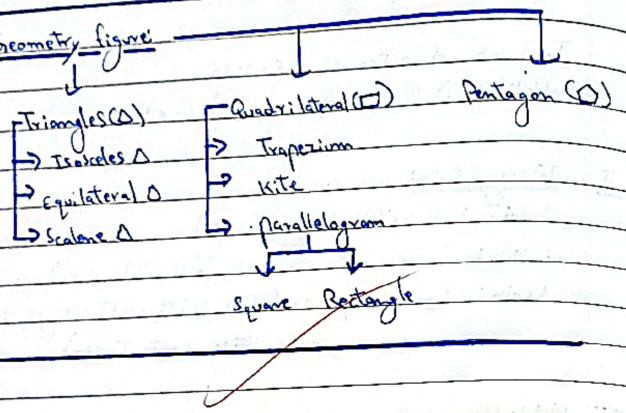
Total no. of cards = 52
Total no. of face cards = 12
Total no. of non-face cards = 40

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Case IV Numbers



Case V Geometry figure:



Exercise 14.1

1-i-Ans Probability of an event E + Probability of the event 'not E' = 1

ii-Ans 0, impossible event

iii-Ans 1, Sure event

iv-Ans 1

v-Ans 0, 1

2-i-Ans When a driver attempts to star, the car starts normally. The car does not start only when there is a defect. Hence, the outcome is not equally likely.

ii-Ans When a player attempts to shoot a basketball the outcomes is not equally likely because the outcomes depends on many factors viz: the training of the player, hieght of the player, etc.

iii-Ans For a question, there are two possibilieties either right or wrong, one out of the two both have equal chances to happen. Hence, the outcomes are equally likely.

iv-Ans A new born baby can either be a boy or a girl. Hence, both the outcomes are equally likely.

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4- Ans Since $0 \le P \le 1$,
$\therefore P(A) = 1.5$ is not possible.

5- Solu we know $P(E) + P(\bar{E}) = 1$
$P(\bar{E}) = 1 - P(E)$
$= 1 - 0.05$
$= 0.95$

6- Solu (i) Since, the bag contains only lemon flavoured candies,
$\therefore$ There is no orange candy.
Hence, the probability of getting an orange candy = 0.
(ii) Since, the bag contains only lemon flavoured candies,
Hence, the probability of getting a lemon flavoured candy = 1.

7- Solu Let the given E be: Two students have the same birthday.
$\therefore$ The event $\bar{E}$ is: Two students do not have the same birthday
Hence, $P(\bar{E}) = 0.992$
$P(E) = 1 - P(\bar{E})$ [$\because P(E) + P(\bar{E}) = 1$]
$= 1 - 0.992$
$= 0.008$

Hence, the probability that two students have same birthday = 0.008.

9- Solu Total number of marbles = 5 red + 8 white + 4 green
= 17
(i) Probability of red ball
$= \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}} = \frac{5}{17}$

(ii) Probability of white ball
$= \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}} = \frac{8}{17}$

(iii) Probability of green ball
$= \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}} = \frac{4}{17}$

12- i- Ans Total number of outcomes = [1, 2, 3, 4, 5, 6, 7, 8]
= 8
Probability of getting 8 = $\frac{1}{8}$

ii- Ans Odd numbers are [1, 3, 5, 7] = 4
$\therefore$ Probability of getting an odd number = $\frac{4}{8} = \frac{1}{2}$

Hence, P(an odd number) = $\frac{1}{2}$

iii- Ans No. greater than 2 are [3, 4, 5, 6, 7, 8] = 6
$\therefore$ Probability of getting a no. greater than 2
$= \frac{6}{8} = \frac{3}{4}$

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13-Sol. When a die is thrown, the possible outcomes are {1, 2, 3, 4, 5, 6} = 6

(i) Prime no. = {2, 3, 5} = 3

Hence, Probability = $\frac{3}{6} = \frac{1}{2}$

(ii) Numbers lying between 2 and 6 = {3, 4, 5}

Hence, Probability = $\frac{3}{6} = \frac{1}{2}$

(iii) Odd no. = {1, 3, 5} = 3

Hence, P = $\frac{3}{6} = \frac{1}{2}$

14-Sol. There are 52 cards in a pack

(i) There are two red kings viz. king of hearts and king of diamonds.

$\therefore P = \frac{2}{52} = \frac{1}{26}$

(ii) There are 12 face cards viz. 4 jacks, 4 queens and 4 kings.

$\therefore P = \frac{12}{52} = \frac{3}{13}$

(iii) There are 6 red face cards viz. 2 jacks, 2 queens, 2 kings.

$\therefore P = \frac{6}{52} = \frac{3}{26}$

(iv) There is only one jack of hearts.

$\therefore P = \frac{1}{52}$

(v) There are 13 spade cards.

$\therefore P = \frac{13}{52} = \frac{1}{4}$

(vi) There is only one queen of diamonds.

$\therefore P = \frac{1}{52}$

17-Sol. (i) Total no. of bulbs = 4 + 16 = 20

No. of defective bulbs = 4

No. of good bulbs = 20 − 4 = 16

Hence, P = $\frac{4}{20} = \frac{1}{5}$

(ii) When a non-defective bulb is drawn,

total no. of bulbs left = 20 − 1 = 19

No. of good bulbs = 16 − 1 = 15

Hence, P = $\frac{15}{19}$

19-Sol. No. of faces of a die = {A, B, C, D, E, A} = 6

(i) There are two A's.

Hence, P = $\frac{2}{6} = \frac{1}{3}$

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(ii) There is only one D.

Hence, $P = \frac{1}{6}$

21-Sol. Total no. of pens in the lot = 144

no. of defective pens = 20

∴ no. of good pens = 144 − 20 = 124

(i) Let the event be E: "She buys a pen".

∴ Hence, Probability that she buys a pen = $\frac{124}{144} = \frac{31}{36}$

(ii) The event $\bar{E}$ is 'She will not buy a pen'.

$P(\bar{E}) = 1 - P(E)$

$= 1 - \frac{31}{36} = \frac{36-31}{36} = \frac{5}{36}$

22-Sol. (I) Let A be the event: "Getting a sum 3".

$A = \{(1,2), (2,1)\}$

$n(A) = 2$

$P(A) = \frac{2}{36} = \frac{1}{18}$

(II) Let B be the event: "Getting a sum 4"

$B = \{(1,3), (3,1), (2,2)\}$

$n(B) = 3$

$P(B) = \frac{3}{36} = \frac{1}{12}$

(III) Let C be the event: "Getting a sum 5"

$C = \{(1,4), (4,1), (2,3), (3,2)\}$

$n(C) = 4$

$P = \frac{4}{36} = \frac{1}{9}$

(IV) Let D be the event: "Getting a sum 6"

$D = \{(1,5), (5,1), (3,3), (2,4), (4,2)\}$

$= 5$

$P = \frac{5}{36}$

(V) Let E be the event: "Getting a sum 7"

$E = \{(1,6), (6,1), (3,4), (4,3), (5,2), (2,5)\}$

$= 6$

$P = \frac{6}{36} = \frac{1}{6}$

(VI) Let F be the event: "Getting a sum 9"

$F = \{(3,6), (6,3), (5,4), (4,5)\} = 6$

$P = \frac{4}{36} = \frac{1}{9}$

(VII) Let G be the event: "Getting a sum 10"

$G = \{(6,4), (4,6), (5,5)\}$

$P = \frac{3}{36} = \frac{1}{12}$

(VIII) Let H be the event: "Getting a sum 11".

$H = \{(5,6), (6,5)\}$

$P = \frac{2}{36} = \frac{1}{18}$

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Event 'Sum of 2 Dice'	Probability
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36

Here's the complete Table

23-Sol. When a coin is tossed three times, then possible outcomes are:

S = {HHH, HHT, HTH, THH, THT, TTH, TTT},

where H = Head and T = Tail.

∴ Total no. of outcomes = 8

Let E be the event: "getting all three same results".

i.e., {HHH, TTT}

$$P(E) = \frac{2}{8} = \frac{1}{4}$$

and $P(\bar{E}) = 1 - P(E) = 1 - \frac{1}{4} = \frac{3}{4}$

Hence, the probability of losing the game is $\frac{3}{4}$.

NCERT Examples

Example 6:-

Solution: Out of the two friends, one girl, say, Savita's birthday can be any day of the year. Now, Hamida's Birthday can also be any day of 365 days ~~in~~ in the year.

we assume that these 365 outcomes are equally likely.

(i) If Hamida's birthday is different from Savita's, the no. of favourable outcomes for her birthday is

365 − 1 = 364

So, $P = \frac{364}{365}$

(ii) ~~P = 1 − P~~

P(Savita and Hamida have the same Birthday)

= 1 − P(both have different birthdays)

$$= 1 - \frac{364}{365} = \frac{365 - 364}{365} = \frac{1}{365}$$

Example 9:-

Solution: Total no. of outcomes = {HH, HT, TH, TT}

= 4

favourable outcomes = {HH, HT, TH} = 3

Probability of getting at least one head =

$$\frac{3}{4}$$